CENG 252 Practice Midterm 1: Solutions

Lectures 1–5 • 100 points • 80 minutes

Problem	Final answers	Similar to
1	(a) $-r_B = 3$, $r_C = 9$; (b) $\mathrm{dm^6/(mol^2{\cdot}s)}$; (c) $E = 12{,}800$ cal/mol, factor 3.83; (d) equal; CSTR larger; (e) $\upsilon/\upsilon_0 = 1.5$, $C_A = C_{A0}/3$	L2, L3, HW1-2
2	(b) 32.0; (c) 31.4; (d) CSTR to 0.6 then PFR: 17.2 $\mathrm{dm^3}$	HW1-1, L2
3	$\varepsilon = 0.5$; $X_e = 0.601$ (flow), 0.556 (batch); $V = 237$ $\mathrm{dm^3}$, $\tau = 4.75$ min	HW1-4(c), L3
4	$X = 0.726$, $y = 0.707$ (0.777 no ΔP); $W_{\max} = 500$ kg, $X_{\max} = 0.865$; $\alpha \times 4$ ($W_{\max} = 125 < 250$ kg); $D_p \times 4$, $X = 0.476$	HW2-1, HW2-2, L4
5	$T(0.8) = 377.1$ K; $V_{\mathrm{CSTR}} = 327$ $\mathrm{dm^3}$; $V_{\mathrm{PFR}} = 334$ $\mathrm{dm^3}$; error 5.7 K	HW2-3, L4/L5

Problem 1: Short answers (15 pts)

(a) Relative rates

For $\mathrm{aA + bB \longrightarrow cC}$, the relative rates are (Lec 3)

$$\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} \quad \Rightarrow \quad \frac{r_A}{-2} = \frac{r_B}{-1} = \frac{r_C}{3}.$$

With $r_A = -6$ mol/(dm^3·s): $\dfrac{-6}{-2} = 3$, so $r_B = -3$ and $r_C = +9$.

Answer

$-r_B = 3$ mol/(dm^3·s) (B is consumed half as fast as A); $r_C = 9$ mol/(dm^3·s).

(b) Units of k

$k = \dfrac{-r_A}{C_A^2 C_B}$, so its units are $\dfrac{\mathrm{mol/(dm^3 \cdot s)}}{(\mathrm{mol/dm^3})^3} = \dfrac{\mathrm{dm^6}}{\mathrm{mol^2 \cdot s}}$. General rule: for overall order n, k has units $(\mathrm{dm^3/mol})^{n-1}/\mathrm{s}$.

Answer

k in $\mathrm{dm^6/(mol^2{\cdot}s)}$.

(c) Activation energy from two temperatures

Step 1. Two-temperature Arrhenius form (Lec 2): $\ln \dfrac{k_2}{k_1} = \dfrac{E}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$.

Step 2. With $k_2/k_1 = 2$:

$$E = \frac{R \ln 2}{\frac{1}{300} - \frac{1}{310}} = \frac{(1.987)(0.6931)}{1.0753 \times 10^{-4}} = 12{,}810 \text{ cal/mol } (= 53.6 \text{ kJ/mol}).$$

Step 3. From 300 K to 320 K: $\dfrac{E}{R} = \dfrac{12{,}810}{1.987} = 6446$ K, and

$$\frac{k_{320}}{k_{300}} = \exp\left[6446\left(\frac{1}{300} - \frac{1}{320}\right)\right] = \exp(1.343) = 3.83.$$

Answer

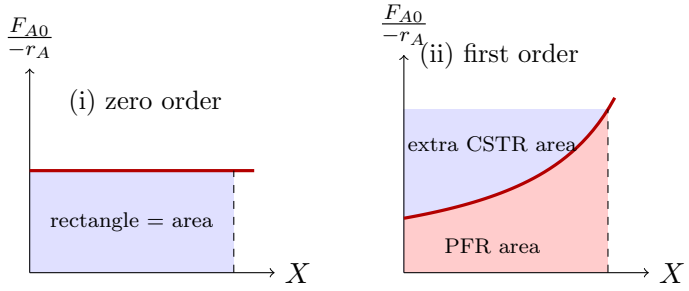
$E \approx 12{,}800$ cal/mol. k rises by a factor of **3.83, not exactly 4**.

Common mistake

"Doubles every 10 K" is only a rule of thumb. The second 10 K step (310→320 K) gives a factor of $3.83/2 = 1.92$, slightly less than 2, because $1/T$ changes less at higher T.

(d) CSTR vs PFR

Design equations: $V_{\text{CSTR}} = \left.\dfrac{F_{A0}}{-r_A}\right|_{\text{exit}} X$ (a rectangle on the Levenspiel plot), $V_{\text{PFR}} = \displaystyle\int_0^X \frac{F_{A0}}{-r_A} dX$ (the area under the curve).



Answer

(i) Zero order: $-r_A = k$ is independent of X, so the curve is a horizontal line and the rectangle equals the area: $V_{\text{CSTR}} = V_{\text{PFR}}$.
(ii) First order: $F_{A0}/(-r_A)$ increases with X. The CSTR runs entirely at the low exit concentration (the highest point on the curve), so $V_{\text{CSTR}} > V_{\text{PFR}}$.

(e) Gas-phase volume change

Step 1. $\delta = 2 - 1 = 1$ and $y_{A0} = 1$, so $\varepsilon = y_{A0}\delta = 1$.

Step 2. Isothermal, no ΔP: $v = v_0(1 + \varepsilon X)\dfrac{P_0}{P}\dfrac{T}{T_0} = v_0(1 + X)$. At $X = 0.5$: $v/v_0 = 1.5$.

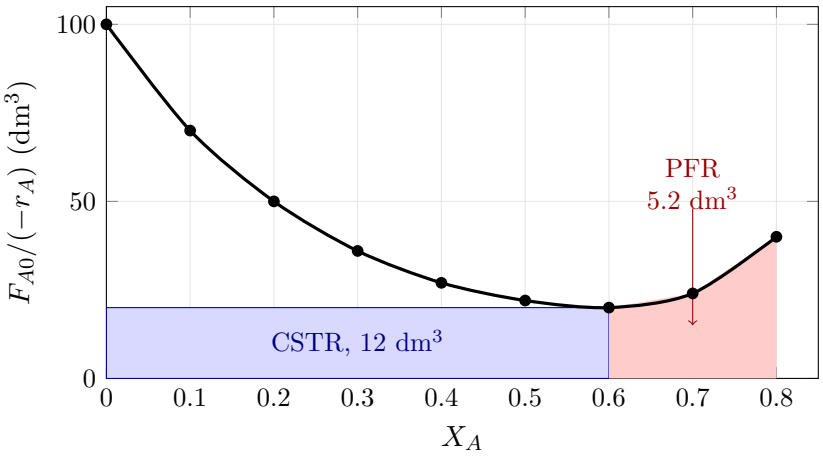
Step 3. $C_A = \dfrac{F_A}{\upsilon} = \dfrac{F_{A0}(1-X)}{\upsilon_0(1+\varepsilon X)} = C_{A0}\dfrac{1-X}{1+\varepsilon X} = C_{A0}\dfrac{0.5}{1.5}$.

Answer

$\upsilon = 1.5\,\upsilon_0$ and $C_A = C_{A0}/3$ (not $C_{A0}/2$: the gas has also expanded).

Problem 2: Levenspiel plot and reactor arrangement (20 pts)

(a) Shape of the curve



Answer

The reaction is exothermic and adiabatic, so T rises with X (energy balance). At low X the temperature increase makes k grow faster (Arrhenius, exponential) than C_A falls, so $-r_A$ increases and $F_{A0}/(-r_A)$ *decreases*. Near the end, reactant depletion dominates, $-r_A$ falls, and the curve turns back up. The minimum is at $X = 0.6$.

(b) Single CSTR to $X = 0.8$

$$V = \left(\frac{F_{A0}}{-r_A}\right)_{X=0.8} X = (40)(0.8).$$

Answer

$V_{\text{CSTR}} = 32.0$ dm^3.

(c) Single PFR to $X = 0.8$

Nine points, $h = \Delta X = 0.1$ (an even number of intervals, so Simpson's 1/3 rule applies):

$$V = \frac{h}{3}\Big[f_0 + 4(f_1 + f_3 + f_5 + f_7) + 2(f_2 + f_4 + f_6) + f_8\Big]$$

$$= \frac{0.1}{3}\Big[100 + 4(70 + 36 + 22 + 24) + 2(50 + 27 + 20) + 40\Big] = \frac{0.1}{3}\big[100 + 608 + 194 + 40\big] = \frac{0.1}{3}(942).$$

Answer

$V_{\text{PFR}} = 31.4$ dm^3, which is almost the same as the CSTR.

(d) Best CSTR + PFR combination

Key idea

Choose the reactor that gives the smaller area on each part of the curve:

- Where the curve *falls* ($0 \to 0.6$), a CSTR's rectangle sits at the *low* exit value, so it lies *below* the curve. The CSTR wins.
- Where the curve *rises* ($0.6 \to 0.8$), the area under the curve is smaller. The PFR wins.

So: **CSTR first, exiting at the minimum ($X_1 = 0.6$), then a PFR to 0.8.**

Step 1. CSTR, $0 \to 0.6$: $V_1 = (20)(0.6 - 0) = 12.0$ dm^3.

Step 2. PFR, $0.6 \to 0.8$, three-point Simpson with $h = 0.1$:

$$V_2 = \frac{0.1}{3}\,[20 + 4(24) + 40] = \frac{0.1}{3}(156) = 5.2 \text{ dm}^3.$$

Step 3. Total: $V_1 + V_2 = 17.2$ dm^3.

Check the other order (PFR first, then CSTR) at the best split: any CSTR that ends at $X = 0.8$ has height 40, so a CSTR from $0.6 \to 0.8$ alone is $(40)(0.2) = 8$ dm^3, plus the PFR $0 \to 0.6$ of 26.2 dm^3, giving 34.2 dm^3. That is much worse.

Answer

CSTR to $X = 0.6$ (12.0 dm^3) followed by a PFR to $X = 0.8$ (5.2 dm^3): **17.2 dm^3 total**, about 46% less than a single CSTR (32.0) or a single PFR (31.4).

Grading guide

(d): 3 pts for the correct order and the reasoning based on the curve's shape, 2 pts for $X_1 = 0.6$, 2 pts for the volumes. A trapezoid rule for the PFR part (5.4 dm^3) gets full credit.

Problem 3: Gas-phase reversible reaction (20 pts)

(a) Stoichiometric table, C_{A0}, ε

Basis: A. $\Theta_B = 0$, $\Theta_I = y_{I0}/y_{A0} = 1$.

Species	Feed (mol/min)	Change	Exit
A	F_{A0}	$-F_{A0}X$	$F_A = F_{A0}(1-X)$
B	0	$+2F_{A0}X$	$F_B = 2F_{A0}X$
I	$F_{I0} = F_{A0}$	0	$F_I = F_{A0}$
Total	$F_{T0} = 2F_{A0}$	$+F_{A0}X$	$F_T = F_{T0} + F_{A0}X = F_{T0}(1+\varepsilon X)$

Step 1. $C_{A0} = \dfrac{y_{A0}P_0}{RT_0} = \dfrac{(0.5)(2)}{(0.08206)(340)} = 0.03584$ mol/dm^3.

Step 2. $\delta = 2 - 1 = 1$ and $\varepsilon = y_{A0}\delta = (0.5)(1) = 0.5$. (Check: F_T/F_{T0} at $X = 1$ is $3/2 = 1+\varepsilon$.)

Step 3. Isothermal, constant P: $\upsilon = \upsilon_0(1+\varepsilon X)$, so

$$C_A = C_{A0}\frac{1-X}{1+0.5X}, \qquad C_B = \frac{2C_{A0}X}{1+0.5X}.$$

Common mistake

$\varepsilon = y_{A0}\delta$ uses the mole fraction of A in the *total* feed, including inerts. Writing $\varepsilon = 1$ (as if A were pure) is the most common error here.

(b) Equilibrium conversion: flow reactor

Step 1. At equilibrium $-r_A = 0$, so $K_C = \dfrac{C_{Be}^2}{C_{Ae}}$:

$$K_C = \frac{4C_{A0}^2X_e^2/(1+\varepsilon X_e)^2}{C_{A0}(1-X_e)/(1+\varepsilon X_e)} = \frac{4C_{A0}X_e^2}{(1-X_e)(1+\varepsilon X_e)}.$$

Step 2. Expand: $K_C(1-X_e)(1+\varepsilon X_e) = K_C\left[1+(\varepsilon-1)X_e - \varepsilon X_e^2\right] = 4C_{A0}X_e^2$. Collect terms:

$$(4C_{A0} + \varepsilon K_C)X_e^2 + K_C(1-\varepsilon)X_e - K_C = 0$$

$$(0.14337 + 0.05)X_e^2 + (0.1)(0.5)X_e - 0.1 = 0 \;\Rightarrow\; 0.19337X_e^2 + 0.05X_e - 0.1 = 0.$$

Step 3. Quadratic formula (positive root):

$$X_e = \frac{-0.05 + \sqrt{0.0025 + 4(0.19337)(0.1)}}{2(0.19337)} = \frac{-0.05 + \sqrt{0.07985}}{0.38674} = \frac{-0.05 + 0.28258}{0.38674}.$$

Answer

$X_e = 0.601$ (flow reactor, constant P).

(c) Equilibrium conversion: constant-volume batch reactor

Step 1. Constant V: $C_A = C_{A0}(1-X)$ and $C_B = 2C_{A0}X$, with no $(1+\varepsilon X)$ factor.

Step 2. $K_C = \dfrac{4C_{A0}X_e^2}{1-X_e} \Rightarrow 4C_{A0}X_e^2 + K_C X_e - K_C = 0 \Rightarrow 0.14337X_e^2 + 0.1X_e - 0.1 = 0.$

Step 3. $X_e = \dfrac{-0.1 + \sqrt{0.01 + 4(0.14337)(0.1)}}{2(0.14337)} = \dfrac{-0.1 + 0.25950}{0.28674}.$

Answer

$X_e = 0.556$ (batch, constant volume), which is lower than the flow value of 0.601.

Why: the reaction makes 2 moles from 1. At constant P the gas expands, so all concentrations drop and the equilibrium shifts toward the side with more moles (Le Chatelier). In the rigid batch vessel the pressure rises instead, which pushes the equilibrium back toward A. This is the classic $N_2O_4 \rightleftharpoons 2\,NO_2$ problem, F3 Example 3-8 (p. 101–105; pure feed gives $X_{e,\text{flow}} = 0.51$ vs. $X_{e,\text{batch}} = 0.44$) and F5 Example 4-5 (p. 126–129).

(d) CSTR volume for $X = 0.8X_e$

Step 1. Target: $X = 0.8(0.601) = 0.481$. Feed: $F_{A0} = C_{A0}\upsilon_0 = (0.03584)(50) = 1.792$ mol/min.

Step 2. Concentrations at $X = 0.481$: $1 + \varepsilon X = 1.2405$, so

$$C_A = \frac{0.03584(0.519)}{1.2405} = 0.01499, \qquad C_B = \frac{2(0.03584)(0.481)}{1.2405} = 0.02780 \text{ mol/dm}^3.$$

Step 3. Rate: $-r_A = 0.5\left[0.01499 - \dfrac{(0.02780)^2}{0.1}\right] = 0.5\,[0.01499 - 0.00773] = 3.632 \times 10^{-3}$ mol/(dm^3·min).

Step 4. CSTR design equation: $V = \dfrac{F_{A0}X}{-r_A} = \dfrac{(1.792)(0.481)}{3.632 \times 10^{-3}}.$

Answer

$V_{\text{CSTR}} \approx 237$ dm^3 and $\tau = V/\upsilon_0 = 4.75$ min.

Precision note

The rate is a difference of two similar numbers $(0.01499 - 0.00773)$, so rounding X to 0.48 or C_{A0} to 0.036 shifts V by a few percent. Any answer of 230–245 dm^3 is correct if the method is right. Keep 4 significant figures in the intermediate steps.

Problem 4: Packed bed with pressure drop (25 pts)

(a) Derivation

Step 1: Pressure. For A $\longrightarrow$ B, $\delta = 1 - 1 = 0$, so $\varepsilon = 0$. Isothermal gives $T/T_0 = 1$:

$$\frac{dy}{dW} = -\frac{\alpha}{2y} \quad \Rightarrow \quad 2y\,dy = -\alpha\,dW \quad \Rightarrow \quad \int_1^y 2y\,dy = -\alpha\int_0^W dW \quad \Rightarrow \quad y^2 - 1 = -\alpha W$$

$$\boxed{y = (1 - \alpha W)^{1/2}}$$

Because $\varepsilon = 0$, the y equation does not involve X, so it can be solved first. This is Case 1 on the Lec 4 slides.

Step 2: Mole balance. PBR: $F_{A0}\dfrac{dX}{dW} = -r'_A$.

Step 3: Rate law + stoichiometry. With $\varepsilon = 0$ and isothermal conditions, $C_A = C_{A0}(1-X)\,y$, so $-r'_A = k'C_{A0}(1 - X)\,y$.

Step 4: Combine. Using $F_{A0} = C_{A0}\upsilon_0$:

$$\frac{dX}{dW} = \frac{k'}{\upsilon_0}(1 - X)(1 - \alpha W)^{1/2}.$$

Step 5: Separate and integrate. ($X = 0$ at $W = 0$)

$$\int_0^X \frac{dX}{1 - X} = \frac{k'}{\upsilon_0}\int_0^W (1 - \alpha W)^{1/2} dW.$$

For the right side, substitute $u = 1 - \alpha W$, $dW = -du/\alpha$:

$$\int_0^W (1 - \alpha W)^{1/2} dW = -\frac{1}{\alpha}\int_1^{1-\alpha W} u^{1/2} du = -\frac{1}{\alpha}\cdot\frac{2}{3}\left[u^{3/2}\right]_1^{1-\alpha W} = \frac{2}{3\alpha}\left[1 - (1 - \alpha W)^{3/2}\right].$$

Step 6: Solve for X.

$$\ln\frac{1}{1 - X} = \frac{2k'}{3\alpha\upsilon_0}\left[1 - (1 - \alpha W)^{3/2}\right] \quad \Rightarrow \quad \boxed{X = 1 - \exp\left\{-\frac{2k'}{3\alpha\upsilon_0}\left[1 - (1 - \alpha W)^{3/2}\right]\right\}}$$

Check: as $\alpha \to 0$, $1 - (1 - \alpha W)^{3/2} \approx \frac{3}{2}\alpha W$, which recovers the no-$\Delta P$ result $\ln\frac{1}{1-X} = k'W/\upsilon_0$.

Slide check

Lec 4 slide 9 integrates this first-order case incorrectly (see Lec4_Notes). Use the result above.

(b) Numbers

$$\frac{k'}{\upsilon_0} = \frac{0.06}{10} = 0.006\ \text{kg}^{-1}, \qquad \frac{2}{3\alpha} = \frac{2}{0.006} = 333.3\ \text{kg}, \qquad \alpha W = 0.5.$$

$$1 - (0.5)^{3/2} = 1 - 0.35355 = 0.64645, \qquad \ln\frac{1}{1 - X} = (0.006)(333.3)(0.64645) = 1.2929$$

$$X = 1 - e^{-1.2929} = 1 - 0.2745 = 0.7255, \qquad y = (0.5)^{1/2} = 0.7071.$$

No ΔP: $\ln \frac{1}{1-X} = k'W/v_0 = (0.006)(250) = 1.5$, so $X = 1 - e^{-1.5} = 0.7769$.

Answer

$X = 0.726$ with $y = 0.707$ ($P = 0.707P_0$). Without pressure drop, $X = 0.777$. Pressure drop costs about 5 percentage points of conversion, because the lower pressure lowers C_A.

(c) Maximum catalyst weight and maximum conversion

$y = 0$ when $1 - \alpha W = 0$, so $W_{\max} = 1/\alpha = 500$ kg. There $(1 - \alpha W)^{3/2} = 0$:

$$\ln \frac{1}{1 - X_{\max}} = \frac{2k'}{3\alpha v_0} = (0.006)(333.3) = 2.0 \quad \Rightarrow \quad X_{\max} = 1 - e^{-2} = 0.865.$$

Answer

$W_{\max} = 500$ kg. Even with the bed packed all the way to $W_{\max}$, X can never exceed **0.865**, whereas without ΔP the conversion would approach 1. Adding catalyst runs into diminishing returns: doubling W from 250 to 500 kg raises X only from 0.726 to 0.865.

(d) Doubling the throughput

From the Ergun equation (Lec 4 slides 7 and 16), $\alpha = \dfrac{2\beta_0}{A_c \rho_c (1-\phi) P_0}$ with

$$\beta_0 = \frac{G(1-\phi)}{\rho_0 g_c D_p \phi^3} \left[\frac{150(1-\phi)\mu}{D_p} + 1.75\, G \right].$$

In turbulent flow the $1.75G$ term dominates, so $\beta_0 \propto G^2/D_p$ and $\alpha \propto \dfrac{G^2}{D_p}$ (with ρ_0, P_0, T_0, A_c, ρ_c, ϕ unchanged). Doubling v_0 at the same inlet density doubles the superficial mass velocity $G = \rho_0 v_0 / A_c$:

$$\frac{\alpha_2}{\alpha_1} = \left(\frac{G_2}{G_1} \right)^2 = 4 \quad \Rightarrow \quad \alpha_2 = 0.008 \text{ kg}^{-1}, \qquad W_{\max} = \frac{1}{\alpha_2} = 125 \text{ kg}.$$

Answer

α increases by a factor of **4** (to 0.008 kg^{-1}). The pressure would reach zero after only 125 kg, which is half of the 250 kg bed, so this bed cannot pass the doubled flow. (In laminar flow $\alpha \propto G$, so it would only double.)

(e) Larger particles

Step 1. To keep the same exit $y = (1 - \alpha W)^{1/2}$ with the same W, α must return to 0.002 kg^{-1}. Since $\alpha \propto G^2/D_p$:

$$\frac{\alpha_3}{\alpha_1} = \left(\frac{G_2}{G_1} \right)^2 \frac{D_{p1}}{D_{p3}} = 4 \frac{D_{p1}}{D_{p3}} = 1 \quad \Rightarrow \quad D_{p3} = 4 D_{p1}.$$

Step 2. Now $\alpha = 0.002$ but $k'/v_0 = 0.06/20 = 0.003$ kg^{-1}:

$$\ln \frac{1}{1-X} = (0.003)(333.3)(0.64645) = 0.6464 \quad \Rightarrow \quad X = 1 - e^{-0.6464} = 0.476.$$

Answer

D_p must be increased **4 times**. Then $y = 0.707$ again, and $X = 0.476$.

Trade-off: the residence time is halved, so the conversion falls from 0.726 to 0.476. However, the production rate of B, $F_{A0}X$, rises from $0.726F_{A0}$ to $2(0.476)F_{A0} = 0.952F_{A0}$, a 31% increase. Also, much larger particles can introduce internal diffusion resistance that would lower the effective k' (a topic for later in the course), so the real conversion could be even lower.

Problem 5: Adiabatic liquid-phase reactor (20 pts)

(a) Energy balance $T(X)$

Step 1: Parameters.

$$\sum \Theta_i \widetilde{C}_{pi} = \Theta_A \widetilde{C}_{pA} + \Theta_B \widetilde{C}_{pB} + \Theta_I \widetilde{C}_{pI} = (1)(20) + (1)(20) + (4)(18) = 112 \text{ cal/(mol A} \cdot \text{K)}$$

$$\Delta \widehat{C}_p = \widetilde{C}_{pC} - \widetilde{C}_{pB} - \widetilde{C}_{pA} = 30 - 20 - 20 = -10 \text{ cal/(mol A} \cdot \text{K)}$$

Step 2: Adiabatic EB. (no work, no heat exchange, Lec 4/5):

$$X = \frac{\sum \Theta_i \widetilde{C}_{pi}\,(T - T_0)}{-\left[\Delta H^\circ_{\text{rxn}}(T_R) + \Delta \widehat{C}_p (T - T_R)\right]}.$$

Step 3: Solve for T. Multiply out: $\sum \Theta_i \widetilde{C}_{pi}(T - T_0) = -X\Delta H^\circ_{\text{rxn}}(T_R) - X\Delta \widehat{C}_p T + X\Delta \widehat{C}_p T_R$. Collect the T terms:

$$T\left[\sum \Theta_i \widetilde{C}_{pi} + X\Delta \widehat{C}_p\right] = \sum \Theta_i \widetilde{C}_{pi}\, T_0 - X\Delta H^\circ_{\text{rxn}}(T_R) + X\Delta \widehat{C}_p T_R$$

$$T = \frac{\sum \Theta_i \widetilde{C}_{pi}\, T_0 + X\left[-\Delta H^\circ_{\text{rxn}}(T_R) + \Delta \widehat{C}_p T_R\right]}{\sum \Theta_i \widetilde{C}_{pi} + X\Delta \widehat{C}_p} = \frac{(112)(300) + X\,[10{,}000 + (-10)(298)]}{112 - 10X}$$

$$\boxed{T = \frac{33{,}600 + 7020\,X}{112 - 10X}} \qquad T(0.8) = \frac{33{,}600 + 5616}{104} = 377.1 \text{ K}.$$

Check: $X = 0$ gives $T = 300$ K. ✓

Common mistake

Lec 4/5 slides 32–35 contain sign typos in the sensible-heat term (see Lec4_Notes). A quick test of your sign: an *exothermic* adiabatic reaction must give $T > T_0$.

(b) CSTR

Step 1. The CSTR operates at the exit temperature, 377.1 K:

$$\frac{E}{R} = \frac{12{,}000}{1.987} = 6039 \text{ K}, \qquad k = 0.01 \exp\left[6039\left(\frac{1}{300} - \frac{1}{377.1}\right)\right] = 0.01\, e^{4.115} = 0.6124.$$

Step 2. $-r_A = kC_AC_B = kC_{A0}^2(1-X)^2 = (0.6124)(1)^2(0.2)^2 = 0.02450$ mol/(dm^3·min), with $F_{A0} = C_{A0}\upsilon_0 = 10$ mol/min.

Step 3. $V = \dfrac{F_{A0}X}{-r_A} = \dfrac{(10)(0.8)}{0.02450}$.

Answer

$V_{\text{CSTR}} = 327$ dm^3 ($\tau = 32.7$ min), operating at 377.1 K.

(c) PFR by Simpson's rule

$\dfrac{F_{A0}}{-r_A} = \dfrac{10}{k(T)(1-X)^2}$, with T from (a) and $k = 0.01\exp\left[6039\left(\frac{1}{300} - \frac{1}{T}\right)\right]$:

X	T (K)	$6039\left(\frac{1}{300} - \frac{1}{T}\right)$	k (dm^3/mol·min)	$(1-X)^2$	$F_{A0}/(-r_A)$ (dm^3)
0	300.0	0	0.01000	1.00	1000
0.2	318.2	1.153	0.03166	0.64	493.5
0.4	337.1	2.216	0.09172	0.36	302.9
0.6	356.7	3.201	0.2455	0.16	254.6
0.8	377.1	4.115	0.6124	0.04	408.2

Five-point Simpson, $h = 0.2$:

$$V = \frac{0.2}{3}\Big[1000 + 4(493.5) + 2(302.9) + 4(254.6) + 408.2\Big] = \frac{0.2}{3}(5006.4).$$

Answer

$V_{\text{PFR}} \approx 334$ dm^3. (An accurate numerical integration gives 331 dm^3, so the five-point Simpson result is within 1%.)

(d) Interpretation

Answer

The **CSTR (327 dm^3) is smaller** than the PFR (334 dm^3). The $F_{A0}/(-r_A)$ column falls from 1000 to 255 and then rises to 408: it has a minimum, just like Problem 2. The PFR has to pass through the cold, slow inlet region (large $F_{A0}/(-r_A)$ near $X = 0$). The CSTR operates entirely at the hot exit condition, where the rate is fast. For an isothermal positive-order reaction the curve only rises, and the PFR is always smaller.

With $\Delta\widehat{C}_p = 0$ you would compute $T = 300 + \dfrac{10{,}000(0.8)}{112} = 371.4$ K, which is **5.7 K too low**. Then $k = 0.480$ instead of 0.612, and the predicted CSTR volume would be 417 dm^3 instead of 327 dm^3 (a 28% error). Because k depends exponentially on T, a few kelvin matter.

At the reactor temperature, $\Delta H_{\text{rxn}}(377\text{ K}) = -10{,}000 + (-10)(377.1 - 298) = -10{,}791$ cal/mol. A less negative $\Delta\widehat{C}_p$ would shrink this correction.

Going further (not graded): as in Problem 2, the smallest total volume would come from a CSTR exiting near the minimum ($X \approx 0.6$), followed by a PFR from $0.6 \to 0.8$.

Grading guide

(a) 2 pts for the EB setup with $\Delta\widehat{C}_p$, 2 pts for the algebra, 1 pt for $T(0.8)$. (c) 3 pts for the table, 3 pts for the Simpson weights, 1 pt for the result. Small differences from rounding T are expected: anything within ± 5 dm^3 of 334 gets full credit.